November 6, 2025

Directive Number: 3B

Group: 1

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As of November 6^{th} , satisfactory progress has been made on the Group Data Project. There have been no changes to the data source or objectives of the selected models. Data cleaning was completed 2 weeks ago, model testing/training is also completed, evaluation is near complete. Next week, we will begin working on preparing visualizations for the data.

The following steps have been completed:

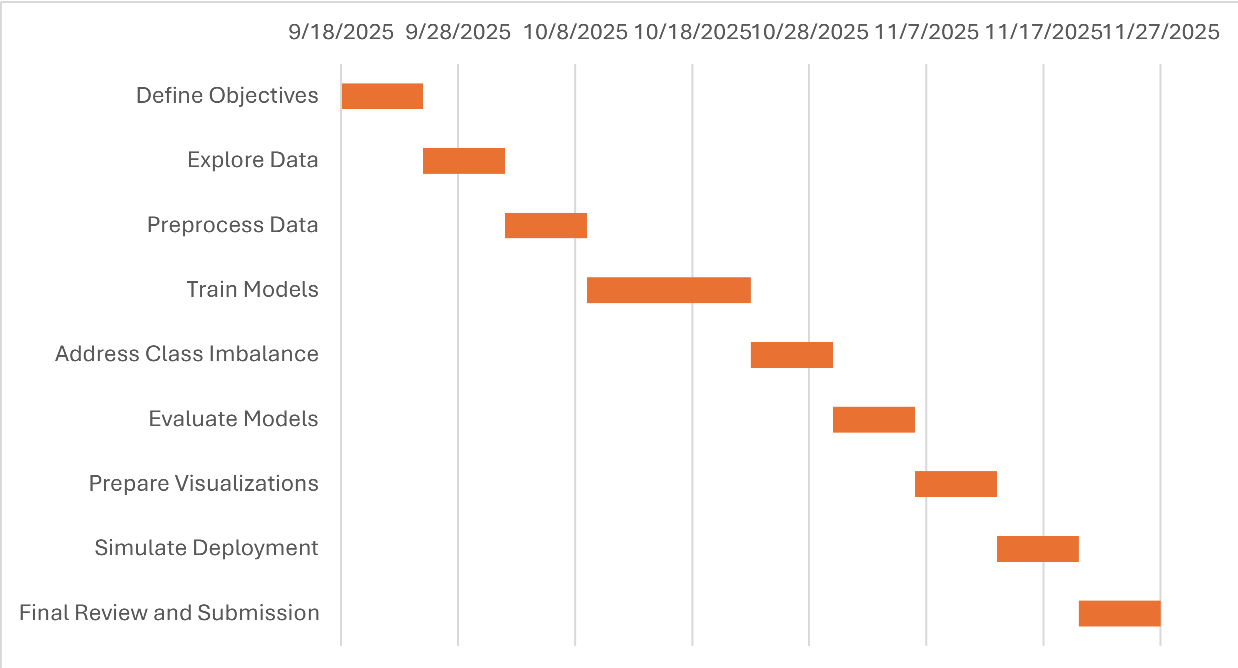
1. Data Integration
    a. Left join train_transaction and train identity on transactionID
    b. drop columns with > 95% missing values
2. Outlier and Zero transaction handling
    a. remove transactions with TransactionAmt == 0
    b. flag top 1% transaction amounts
3. Missing Value Treatment
    a. Numerical Features
        i. impute per class (fraud vs legit) using median
        ii. Create binary feature_missing indicator for columns with <20% missing
    b. Categorical Features
        i. If <95% missing, impute with "missing"
4. Encoding
    a. Label encoding for high-cardinality categorical features
    b. One-hot encoding for low-cardinality categorical features
5. Feature Engineering
    a. Temporal Features
        i. Derived from TransactionDT to capture time-of-day, day-of-week, and transaction frequency patterns
    b. User Level Aggregates
        i. Aggregated transaction-level data at the card1 (user) level. Created summary statistics per user, including mean and standard deviation of TransactionAmt, 7-day rolling transaction counts, 30-day rolling standard deviation of amounts, and count of unique devices over the past 30 days. Missing standard deviations imputed with 0.
    c. Log Transformation
        i. Applied log(1 + x) to reduce skewness and stabilize variance

6. Feature Selection
    a. Remove highly correlated features
        i. Dropped redundant numeric columns based on high pairwise correlation to reduce multicollinearity and simplify model training. Went from 893 features to 331 features with this step
7. Handling Class Imbalance
    a. SMOTE was skipped because it cannot reliably process very large, high-dimensional datasets with extreme class imbalance, even when there are no missing values. Instead, class imbalance was handled directly in XGBoost using scale_pos_weight.
8. Output
    a. Export datasets for future use
        i. Saves entire datasets as RDS for reproducibility and sharing. Retains column types and structure. Can be loaded with readRDS() in future sessions or by collaborators.
9. Model Testing
    a. Tested Logistic Regression, Random Forest, and XGBoost
10. Model Training
    a. Cross-validation to optimize model performance
11. Evaluation (NEAR COMPLETE)
    a. Seeking accuracy, precision and performance
    b. XGBoost model provides the most balance of all 3

Current Goals:

- Develop strong visualizations that convey a message to the audience
- Successfully deploy the model and use as intended
- Integrate all visuals and models into a comprehensive final report

Gantt Chart:



When viewing our Gantt chart, it is clear that we are right on schedule. Currently, our team is near completion in the evaluating models phase. This puts us on track to complete the project on time, as next week we will begin to prepare visualizations. The only minor deviation occurred during data cleaning; additional time was spent fine-tuning the details. However, this adjustment improved the data quality and did not delay further deadlines.

The group has maintained consistent communication, efficient task completion and high quality of work. With evaluation almost complete and visuals underway, the project remains well-positioned for timely and successful completion.